

Dhruv Nirmal

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Summary

Data & ML Engineer with 2+ years delivering production ETL pipelines, ML models, and data-quality systems on large enterprise datasets. Built 20+ automated data pipelines processing millions of transactions monthly; delivered ML models (classification, forecasting, anomaly detection) and agentic AI workflows at production scale. Strong Python, SQL, Databricks (PySpark), Docker, and AWS experience with a proven track record across the full data-to-insight stack — from ingestion pipelines to BI dashboards. Passionate about building reliable data infrastructure that enables ML delivery and business decision-making at scale.

Experience

Data Engineer — Purchasing Index Data Analytics (Comprara Group) Jun 2024 – Present

Melbourne, Australia

- Built and maintained 20+ automated ETL pipelines processing millions of procurement transactions monthly across 5 enterprise clients; end-to-end pipeline ownership from data ingestion through transformation to reporting delivery (Python, SQL, Azure-hosted infrastructure).
- Built a multi-agent data QA validation workflow (5 agents + orchestrator) to validate and enforce data integrity on client datasets prior to delivery — caught structural and value anomalies before reaching stakeholders.
- Developed a binary classification ML model (TF-IDF + Logistic Regression) to categorise financial transaction descriptions, cutting manual effort ~40% and improving data standardisation for downstream analytics pipelines.
- Built an ML model to predict chemical consumption for industrial clients using forecasting techniques, improving production planning and inventory management decisions.
- Performed anomaly detection on procurement datasets, applying statistical analysis to detect cost-leakage patterns — contributed to ~15% client savings.
- Developed and maintained 16+ Tableau dashboards for procurement and finance stakeholder teams; communicated complex analytical outputs as actionable business insights for senior executives.
- Built agentic AI workflows using Claude to auto-generate structured client reports and convert analytics outputs to PowerPoint decks — eliminated manual assembly and standardised delivery.
- Integrated LLM tools (Claude, ChatGPT) into analytics script development, accelerating pipeline delivery and reducing manual coding effort.

Data Engineer Intern — Victorian Centre for Data Insights (VCDI) Aug 2023 – Nov 2023

Melbourne, Australia

- Built a distributed anomaly-detection pipeline in Databricks using PySpark to process government procurement data at scale; improved anomaly detection accuracy ~20%.
- Engineered scalable data transformation layers in cloud environments for large-scale procurement datasets — responsible for ingestion, cleansing, and serving-layer delivery.
- Deployed a Power BI analytics solution adopted by senior Department of Transport stakeholders; presented system architecture and model outcomes to cross-functional teams.

Research Data Analyst — Terminal Ballistics Research Laboratory (TBRL, DRDO) Jan 2021 – Jul 2021

- Analysed blast-wave data with statistical models; applied Butterworth filter to reduce noise ~20%; improved prediction accuracy ~20% and defence equipment design efficiency ~10%.

Projects

Cloud Data Platform with Terraform (Personal)

- Provisioned full AWS cloud infrastructure using Terraform IaC (S3, IAM, EC2, networking) for reproducible multi-environment deployments (dev/staging/prod) — containerised using Docker.
- Built an automated data ingestion pipeline from external APIs to Snowflake; end-to-end data platform from ingestion to BI reporting (Power BI).

Education

Master of Data Science — Monash University Feb 2022 – Dec 2023

Machine Learning, Statistics, Applied Forecasting, High Dimensional Analysis, Supply Chain Management, Web App Development

Bachelor of Engineering — Thapar University May 2017 – May 2021

Skills

Data Engineering: Python, SQL, ETL/ELT pipeline design, data modelling, data quality & integrity, anomaly detection, KNIME, Snowflake, feature engineering

Cloud & Infrastructure: AWS (S3, Lambda, IAM, EC2), Azure, Databricks, Docker, Terraform (IaC); picking up: GCP (BigQuery, Cloud Functions)

Machine Learning & AI: scikit-learn, TF-IDF, Logistic Regression, forecasting models, PySpark, LLM/agent workflows (Claude, ChatGPT), multi-agent orchestration, NLP/sentiment analysis, anomaly detection

Pipeline & Orchestration: Batch pipeline automation, Python scripting, scheduled workflow automation; actively upskilling: Apache Airflow, dbt

BI & Visualisation: Tableau (16+ production dashboards), Power BI, KPI analytics, data storytelling